

EMMANUEL M NDALIRO

Software Engineer | Python &
Machine Learning

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PROFILE SUMMARY

Results-driven Software Engineer with 6+ years of experience, specializing in Python and advanced Machine Learning. Demonstrated success in architecting, training, fine-tuning, and deploying Large Language Models (LLMs) and AI-powered applications, including Retrieval-Augmented Generation (RAG), Agentic RAG, AI agents, and protocol-driven solutions (MCPs & A2A). Hands-on experience with local AI development using Ollama and relevant open-source tools. Strong backend engineering skills, quality assurance, and the ability to validate and optimize LLM outputs with real-world datasets for production environments.

EDUCATION

Multimedia University of Kenya

BSc. in Software Engineering

- From 2015 September to 2019 October. - Graduated with 2nd class honors.

St. Joseph's Boys School Kitale

Graduated with KCSE certificate 2014

- Excellence with A- mean grade.

CAREER BACKGROUND

Python, Machine Learning & AI Development (2025)

- Built and deployed machine learning models and LLM-powered applications focused on real-world language understanding, retrieval-augmented generation (RAG), and agentic workflows.
- Hands-on experience training and fine-tuning language models (LLMs) and deploying models with frameworks such as Huggingface Transformers and local AI solutions (including Ollama).
- Developed sophisticated AI applications featuring LLM inference, RAG, Agentic RAG, Hybrid RAG, AI agent protocols (MCPs & A2A), and LLM chaining for advanced automation.
- Designed and implemented validation pipelines to evaluate LLM output, provide model feedback, and enhance response quality using open-source datasets.
- Integrated AI/ML services with secure Python backends (Flask, Django) and scalable cloud infrastructures (AWS, Azure, GCP).
- Built Flutter and web frontends consuming AI-powered APIs for automated, intelligent user experiences.
- Automated data flows, QA, and business processes using agentic AI, Python scripting, and cloud functions.
- Collaborated in Agile teams to deliver production-grade AI and machine learning features on-time and at scale.

Python Backend, Flutter and Data Science Path

From July 2023 to December 2024.

- Orchestrated Flutter and Dart solutions for diverse projects for cross-platform applications.
- Automated processes utilizing Python in conjunction with UIPath, Selenium and the PlayWright Framework

- Automating Business Processes with LLMs” and “Runner H & Surfer H: A Masterclass in Modern Browser-Use Agents.”
- Architected next-generation AI agents addressing scalability, interoperability, and private/local AI (“The Ultimate Guide to Local AI and AI Agents: Building Private, Powerful AI Systems”).
- Extensive publication, open-source project contributions, and community engagement.
- Enabled business automation and data-driven workflows using AI agent frameworks.
- Flutter frontend development.

APPLIED MACHINE LEARNING & AGENTIC AI

- Applied Machine Learning & Agentic AI
- Published technical articles on Medium covering agentic AI, Model Context Protocol (MCP), and A2A agent interoperability, such as “The AI Protocol Wars: A Comprehensive Analysis of MCP vs. A2A with Practical Implementations.”
- Built structured AI agents using Python and frameworks like Pydantic and LangGraph, showcased in Medium posts and open-source codebases.
- AI Agents and Multi-Agent Automation
- Researched and documented context engineering and dynamic agent systems (“Agent Intelligence: A Strategic Approach to Context Engineering,” “Building Structured AI Agents: Harnessing DeepSeek R1 and Pydantic AI”).
- Contributed open-source automation tools and agent frameworks on GitHub.
- Web Scraping Automation
- Developed and published on modern AI-powered web scraping, automating information retrieval as part of larger agentic systems.

Python, Machine Learning (2023)

- Advanced Automation and Flutter Integration
 - Increased focus on automation systems and integration with production-level Flutter applications, achieving streamlined cross-platform solutions.
 - Deepened Python expertise, leveraging new Google certification in developer training to enhance and professionalize automation workflows.
- Machine Learning
 - Began incorporating more advanced ML models in projects. Used GitHub for sharing and versioning data science scripts.
- Multi-agent Systems (Emerging)
 - Early exploration of agentic concepts and automation protocols in AI.

January 2022

Python, Machine Learning & AI Development (2022)

- Continued building Python-based automation scripts and tools, laying groundwork for robust machine learning pipelines and reusable code modules.
- Engaged with open-source communities via GitHub, contributing to and maintaining repositories that support automation and data workflows.
- Early Machine Learning Projects
- Developed initial machine learning models for structured data.
- Automation Foundations
- Experimented with test automation (evidenced by completion of “Test Automation University Python Development & Pytest Automation” and Google certificate programs).

Mobile & Web Development. Python Backend and Data Science Path

From July 2021 to December 2022.

- Orchestrated Flutter and Dart solutions for diverse projects for cross-platform applications.
- Automated processes utilizing Python in conjunction with UIPath, Selenium and the PlayWright Framework

TECHNICAL SKILLS

- AI/ML Development: LLM training, fine-tuning & inference; RAG, Agentic RAG, Hybrid RAG; AI agents & MCPs/A2A protocols; LLM chaining/workflows
- Local AI: Ollama, on-premise model deployment & eval
- AI/LLMs Guardrails
- Python Ecosystem: Flask, Django, Pandas, Numpy, Scikit-learn, Huggingface Transformers
- Cloud & APIs: AWS, Azure, GCP, RESTful integration, CI/CD
- Mobile/Web: Flutter (Dart), Firebase, WebSockets
- Backend Development: Python in designing, developing, and maintaining efficient backend systems Flask and Django.
- Restful APIs: Building and using RESTful APIs with Python Flask for smooth data exchange.
- Git and Bitbucket.
- Cloud & DevOps: Familiar with AWS, Azure, Google Cloud, Docker, CI/CD pipelines.

CERTIFICATIONS

- **BSc.** Software Engineering from the Multimedia University of Kenya.
- Foundation for Test Automation (Test Automation University)
- Python Programming (Test Automation University)
- Source Control for Test Automation (Test Automation University)
- Data Science
- Google Africa Developer Training Program

Automated testing by Test Automation University

Backend Developer

SafeTek Limited | From April 2021 to July 2021.

- Managed Python-driven backend development, enhancing product capabilities; Flask and Django
- Oversaw database solutions with Firebase and SQLite, promoting efficient data stewardship.
- Deepened knowledge base through hands-on experience with Robotics Framework and RESTful APIs for optimal data exchange.
- Used Git and Bitbucket for team collaboration and smooth project progression.

Mobile Developer

From May 2020 to March 2021

- Utilized Python for backend mobile application development, contributing to the creation of various high-functioning apps.
- Flutter and Dart for cross-platform and Firebase Database.
- Leveraged Google Maps APIs for location-based features in mobile apps.
- Collaborated with UX designers to convert wireframes and prototypes into functional and aesthetically pleasing designs.

LANGUAGES

- English
- Kiswahili

REFERENCES

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